

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

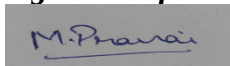
Descriptive Written Test

Code of Conduct:

I M.Pranai (192525068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ANSWER

1. Intelligent Agents in Smart Home Automation

Introduction

A smart home automation system manages lighting, temperature, and security using inputs such as motion, temperature, and time. It can also learn the user's preferences over time. Therefore, different types of intelligent agents can be used to make the system more efficient.

Types of Intelligent Agents

1. Simple Reflex Agent

A simple reflex agent selects an action based only on the **current percept**.

Process:

Percept → Condition → Action

Input	Rule	Action
Motion detected	If motion = ON	Turn ON light
High temperature	If temperature > limit	Turn ON AC
No motion	If no motion	Turn OFF light

Advantages:

- Simple to design.
- Fast response.
- Easy to implement.

Limitation:

- Does not remember past situations.
- Cannot learn user preferences.

2. Model-Based Agent

A model-based agent maintains an **internal state** of the environment.

Process:

Percept → Update Internal State → Decide Action → Perform Action

Example:

If the system knows that a person is currently in the living room and the room was previously dark, it can use this information to control the light.

Feature	Example
Current input	Motion detected
Internal state	Person is in living room
Decision	Turn on living-room light

Advantage: Works better when the environment is not completely observable.

3. Goal-Based Agent

A goal-based agent chooses actions to achieve a specific **goal**.

Possible smart-home goals:

- Maintain comfortable temperature.
- Keep the house secure.
- Reduce unnecessary energy usage.

Process:

Current State → Goal → Select Action → Goal Achieved

Example:

Goal = Maintain comfortable temperature.

If temperature is high:

Detect temperature → Compare with goal → Turn ON AC → Reach comfortable temperature

4. Utility-Based Agent

A utility-based agent chooses the action that gives the **best overall utility**.

It can consider several factors at the same time:

- Comfort

- Energy consumption
- Security
- Cost
- User preferences

For example, instead of simply turning the AC ON whenever temperature increases, it can consider the temperature, user's preference, electricity consumption and time of day before deciding.

Process:

Percepts → Evaluate Alternatives → Calculate Utility → Select Best Action

Comparison

Agent	Main Feature	Smart Home Example
Simple Reflex	Current condition	Motion → Light ON
Model-Based	Uses internal state	Remembers room status
Goal-Based	Achieves goals	Maintain desired temperature
Utility-Based	Selects best option	Comfort + energy saving

Most Effective Approach

A **hybrid approach combining model-based, goal-based and utility-based agents** would be most effective.

Why?

1. **Model-based** agent maintains information about the home.
2. **Goal-based** agent helps achieve comfort and security goals.
3. **Utility-based** agent balances comfort, energy efficiency and security.
4. A learning component can adapt to user preferences over time.

Example

Suppose the user normally prefers lower temperature at night.

The system can:

Detect time → Check temperature → Remember user preference → Compare energy usage → Adjust AC

Conclusion

A **hybrid intelligent agent**, mainly using model-based and utility-based decision-making with goal-based behaviour, is the best choice for a smart home. It can provide **comfort, energy efficiency and security** while adapting to user preferences.

2. Robot in an Unknown Maze – UCS and IDS

Introduction

A robot is placed in an unknown maze and must find a path to the exit without knowing the layout beforehand. Uninformed search strategies such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can be used to explore the maze.

1. Uniform Cost Search (UCS)

UCS expands the node having the **lowest path cost**.

Process

Start → Calculate path cost → Select lowest-cost node → Expand → Repeat → Goal

Suppose:

Path	Cost
S → A	2
S → B	5
A → G	3
B → G	2

UCS compares cumulative costs:

- $S \rightarrow A = 2$
- $S \rightarrow A \rightarrow G = 2 + 3 = 5$
- $S \rightarrow B = 5$
- $S \rightarrow B \rightarrow G = 5 + 2 = 7$

Therefore, UCS selects:

S → A → G

Total cost = 5

Advantages

- Complete when appropriate cost conditions hold.

- Finds the **least-cost path**.
- Useful when different movements have different costs.

Disadvantages

- Can require large memory.
- May explore many nodes.
- Can be slower when the search space is large.

2. Iterative Deepening Search (IDS)

IDS combines the advantages of **Depth-First Search** and **Breadth-First Search**.

It repeatedly performs depth-limited DFS with increasing depth.

Process

Depth 0 → Depth 1 → Depth 2 → Depth 3 → ... → Goal

Example:

Iteration	Depth Limit	Process
1	0	Check Start
2	1	Explore nodes at depth 1
3	2	Explore nodes up to depth 2
4	3	Explore nodes up to depth 3
5	4	Continue until goal is found

Advantages

- Uses relatively low memory.
- Complete for finite branching factor.
- Useful when the depth of the solution is unknown.
- Finds a shallow solution when all step costs are equal.

Disadvantages

- Repeats earlier searches.
- Can take more time due to repeated node expansion.
- Not suitable for different path costs when the objective is minimum cost.

UCS vs IDS

Feature	UCS	IDS
Main idea	Lowest path cost	Increasing depth
Cost consideration	Yes	No
Memory	High	Low
Complete	Yes, under standard positive-cost conditions	Yes for finite branching
Optimal	Yes for positive costs	Optimal for equal step costs
Best use	Different path costs	Unknown solution depth

Intelligent Agent in the Maze

The robot can be considered an **intelligent agent** because it:

Perceives → Decides → Acts

Simple Reflex Agent

Uses the current situation only.

Example:

Obstacle detected → Turn

Model-Based Agent

Maintains information about already explored parts of the maze.

Example:

Remember visited cells → Avoid repeating paths

Goal-Based Agent

Uses the goal of reaching the exit.

Current position → Search path → Exit

Utility-Based Agent

Can consider additional factors such as:

- Distance
- Energy consumption
- Safety
- Time

Most Effective Combination

For an unknown maze where each movement has the **same cost**, **IDS with a model-based/goal-based agent** can be a good choice.

Process

Start



Observe surroundings



Update internal map



Use IDS to search progressively deeper



Avoid already known blocked paths



Reach goal

Why?

- The maze layout is initially unknown.
- The robot needs to remember explored information.
- IDS requires less memory than UCS.
- If every movement has equal cost, the shallowest solution is also a least-step solution.
- A goal-based agent keeps the robot focused on reaching the exit.

If the maze has **different movement costs**, then **UCS with a model-based/goal-based agent** would be preferable because UCS considers actual path cost.